2025 Global Conference in Emerging Technology (GINOTECH)
Pune, India. May 09 - 11, 2025

AfterLife Connect: An AI-Based Platform for Interactive Digital Avatars

Shivam Awasare[1]
Artificial Intelligence and Data Science
Zeal College of Engineering and Research
Pune, India
shivammawasare@gmail.com

Lavanya Moolya[2]
Artificial Intelligence and Data Science
Zeal College of Engineering and Research
Pune, India
lavanyamoolya227@gmail.com

Siddharth Khodke[3]
Artificial Intelligence and Data Science
Zeal College of Engineering and Research
Pune, India
siddharthkhodke1919@gmail.com

Sudhanshu Dixit[4]
Artificial Intelligence and Data Science
Zeal College of Engineering and Research
Pune, India
sudhanshudix27@gmail.com

Prof. V. N. Nayakwadi[5]
Artificial Intelligence and Data Science
Zeal College of Engineering and Research
Pune, India
vishal.nayakwadi@zealeducation.com

2025 Global Conference in Emerging Technology (GINOTECH) | 979-8-3315-0775-6/25/$31.00 ©2025 IEEE | DOI: 10.1109/GINOTECH63460.2025.11076783

***Abstract*—The project titled *AfterLife Connect* is an innovative exploration into using artificial intelligence to create interactive digital avatars. The web-based application enables users to engage in discussions and receive answers that imitate real-sounding speech patterns, conversational styles, and individual personalities. Using advanced natural language processing (NLP), speech synthesis, and image generation technologies, the platform creates an immersive experience, demonstrating the potential of AI in dynamic interaction. The project focuses on using custom-generated datasets to train AI models. It addresses challenges such as data limitations, ethical considerations, and ensuring accurate personality simulation. Ethical safeguards prevent misuse and ensure responsible AI development. Although it currently relies on custom datasets, the project lays the groundwork for future applications that could include recreating avatars of celebrities or loved ones, provided there is accurate enough data. *AfterLife Connect* bridges the gap between static records and interactive AI experiences, offering a novel way to engage with digital personas.**

***Index Terms*—Artificial Intelligence, Natural Language Processing, Voice Synthesis, Digital Avatars, Ethical AI, Interactive Learning**

I. Introduction

The *AfterLife Connect* initiative seeks to transform how individuals engage with digital avatars of personalized personas. Utilizing progress in artificial intelligence (AI), the platform produces virtual avatars that can partake in realistic dialogues, create authentic images and videos, and offer speech synthesis. This initiative offers an engaging experience, allowing users to engage with these virtual avatars, merging AI with human recollection.

The platform uses advanced AI technologies such as natural language processing (NLP), image and video creation, and voice generation to capture the essence of a particular person. By utilizing these technologies, the system produces tailored dialogues, vocal outputs, and images/videos that replicate realistic behavior, looks, and speech patterns informed by unique data. Users can ”interact” with these avatars in real-time, obtaining replies that resemble natural conversation.

From education to entertainment, AfterLife Connect offers a range of potential applications that include mental health support.Virtual objects can be utilized as educational resources, personalized learning experiences for children, or companions for those experiencing emotional distress.Furthermore, the site can be used to archive and replicate certain aspects of people’s pasts and personalities, providing users with a unique opportunity to connect with memories and individuals from the past.The new AI-led design philosophy of AfterLife Connect seeks to revolutionize the way people interact with digital characters, while keeping ethics at the forefront through a revolutionary approach.

A. Challenges Faced During Development

Key challenges included:

- **Processing Power:** Addressing high computational demands.
- **Data Quality:** Thorough preprocessing and augmentation guaranteed precision.
- **Ethical Constraints:** Followed protocols to avoid false representation.
- **Model Optimization:** Iterative fine-tuning adjusts models according to unique characteristics of a custom dataset.
- **Integration Issues:** A strong backend facilitated smooth integration of NLP, voice, and image modules.

Authorized licensed use limited to: Shah and Anchor Kutchhi Engineering College. Downloaded on March 02,2026 at 04:08:10 UTC from IEEE Xplore. Restrictions apply.

B. Project Approach

Our approach includes:

- **Data Preparation:** The web-based application enables users to engage in discussions and receive answers that imitate real-sounding speech patterns, conversational styles, and individual personalities.
- **NLP Models:** Adjusting transformers (GPT-3, Llama 3, Mistral 7B) to generate persona-targeted replies.
- **Voice Synthesis:** Text-to-Speech systems (Tacotron 2, WaveNet, F5 TTS) featuring voice replication for authentic sound
- **Image Generation:** GANs (StyleGAN3, FOMM) generated realistic avatars.
- **Ethical Safeguards:** Added filters to guarantee that AI results were accountable and matched with validated information.
- **System Integration:** Backend APIs linked modules for instant interaction.

II. Literature Review

1. Chatbot for the Educational Domain *Authors*: Anupam Mondal, Monalisa Dey, Dipankar Das, et al.

Mondal et al. [12] proposed a self-contained chatbot with specialized features for the educational sector.The system uses a learning approach and subsequently relies on identifying methods to extract 1000 question-answer pairs from an unstructured educational dataset using npm, noise extraction, and the random forest classifier. Key features such as term frequency, inverse document frequency (TF-IDF), question types, and linguistic cues were used to improve response accuracy.After achieving an F-measure of 0.870, the chatbot was declared a functional Telegram bot and showcased its potential for use in various scenarios.

2. Personality Traits Analysis Using Artificial Neural Networks *Authors*: Ahmed Remaida, Benyoussef Abdellaoui, Aniss Moumen, et al.

Remaida et al. [13] conducted an extensive literature review on the role of artificial neural networks (ANNs) in personality trait analysis. The study examined 125 research articles that used topic modeling techniques, such as latent Dirichlet distribution (LDA). The Big-Five personality model was identified as the most widely used framework. The survey focuses on the role of ANNs in automatic personality recognition, perception, and synthesis, highlighting challenges and trends in processing diverse inputs. such as text, audio, and video for personality prediction.

3. Scripted Video Generation with Bottom-Up GAN *Authors*: Qi Chen, Qi Wu, Jian Chen, et al.

Chen et al. [14] presented a new framework for text-to-video synthesis using bottom-up generative adversarial networks (BoGAN). The model integrates multi-level semantic alignment mechanisms, including discriminators at the region, image, and video levels, to ensure the consistency and fidelity of the generated videos. Tested on synthesis (SBMG, TBMG) and real-world (MSVD, KTH) datasets, BoGAN outperformed state-of-the-art methods using metrics such as the initial Fréchet distance (FID). This method has shown significant potential for applications in video editing, gaming, and educational content creation.

4. AI and Web-Based Human-Like Interactive University Chatbot (UNIBOT) *Authors*: Neelkumar P. Patel, Devangi R. Parikh, Darshan A. Patel, Ronak R. Patel

This study focuses on creating a university chatbot that allows quick access to student information such as tuition fees, calendars, and admission processes. The chatbot is implemented with PHP for the backend and pre-processes user requests to retrieve relevant data. Future enhancements include the integration of natural language processing (NLP) for better interaction.[15]

5. Artificial Intelligence Architecture Inspired by Personality Theory *Authors*: Gabriel A. Popoola, Corey A. Graves

This article presents a classification architecture (JPICA) inspired by Jungian psychology. She applies personality theory to artificial intelligence (AI), particularly to speech segmentation tasks. The architecture combines machine learning models and psychological features, which improve classification accuracy.[16]

6. Vision Powered Conversational AI for Easy Human Dialogue Systems *Authors*: Bipendra Basnyat, Neha Singh, Nirmalya Roy, Aryya Gangopadhyay

This book introduces "FloodBot," a conversational AI designed for flood risk assessment. It integrates computer vision and natural language processing (NLP) models, leveraging transfer learning to analyze flood data and improve query response capabilities. The system has been validated in real-world deployments.[17]

7. Digital Persona Portrayal: Identifying Pluridentity Vulnerabilities in Digital Life *Authors*M. A. Olivero, A. Bertolino, F. J. Domínguez-Mayo, M. J. Escalona, and I. Matteucci

The paper explores the notion of multi-identity, which involves multiple online identities, and the potential risks associated with data manipulation.It presents a metamodel for digital personality representation and an analysis process to identify weaknesses caused by data exposure.Through profiles, the authors expose how attackers can manipulate victims into social engineering or to bypass security measures.[21]

8. Validating Social Media Data for Automatic Persona Generation *Authors*J. An, H. Kwak, and B. J. Jansen

In this article, we explore the concept of multi-identity, where individuals are represented by multiple online identities, and the potential risks associated with data exploitation.The metamodel for digital personality representation and an analysis process to identify weaknesses resulting from data overexposure across these identities are presented.The authors illustrate how attackers can use victim profiles to create social engineering or security loopholes.[**?**]

Authorized licensed use limited to: Shah and Anchor Kutchhi Engineering College. Downloaded on March 02,2026 at 04:08:10 UTC from IEEE Xplore. Restrictions apply.

TABLE I
SUMMARY OF LITERATURE REVIEW

Title	Authors	Methodology
Chatbot: An Automated Conversation System for the Educational Domain	A. Mondal et al.	Random Forest Classifier
Personality Traits Analysis Using Artificial Neural Networks: A Literature Survey	A. Remaida et al.	Literature Survey using LDA Topic Modeling
Scripted Video Generation With a Bottom-Up GAN	Q. Chen et al.	GAN-based Video Generation
AI and Web-Based Human-Like Interactive University Chatbot (UNIBOT)	N. P. Patel et al.	PHP Backend Development
Artificial Intelligence Architecture Inspired by Personality Theory	G. A. Popoola et al.	Jungian-Inspired Classification Architecture
Vision Powered Conversational AI for Easy Human Dialogue Systems	B. Basnyat et al.	NLP and Vision-Based Conversational AI
Digital Persona Portrayal: Identifying Pluridentity Vulnerabilities in Digital Life	M. A. Olivero et al.	Analysis of data leaks and security mechanisms using Digital Persona Portrayal Metamodel
Validating Social Media Data for Automatic Persona Generation	J. An et al.	Statistical analysis of YouTube data for real-time persona generation

III. RATIONALE

A. Reasons for Concern

Conventional approaches, like biographies and documentaries, provide restricted interaction with historical or cultural personalities. *AfterLife Connect* seeks to fill this void by delivering interactive digital avatars created from customized datasets, illustrating how AI can replicate human-like exchanges with personalized identities.

B. Expected Impact

The initiative provides a basis for upcoming educational resources and entertainment software by mimicking realistic interactions. It demonstrates how AI can generate tailored, interactive dialogues, paving the way for future developments, such as simulating well-known figures or cherished individuals when precise data is accessible.

C. Motivation

The main objective is to investigate the convergence of AI and human memory. Utilizing cutting-edge AI technologies, *AfterLife Connect* seeks to showcase the possibilities of realistic, AI-powered avatars, delivering a distinctive, immersive experience rooted in personalized datasets.

IV. OBJECTIVES

The *AfterLife Connect* initiative seeks to accomplish the subsequent goals:

- **Create Interactive Digital Avatars:** Create AI-driven avatars that can engage in authentic text and voice interactions utilizing personalized datasets.
- **Accurate Persona Simulation:** Make certain that the AI mimics particular speech patterns, behaviors, and dialogue styles by utilizing customized data.
- **Generate Realistic Visual Representations:** Utilize GANs to generate realistic avatars that boost user interaction.
- **Seamless Model Integration for Enhanced User Experience:** Combine NLP, voice synthesis, and image generation models into a unified web application, providing an intuitive interface that is accessible to users lacking technical skills.
- **Promote Ethical AI Use:** Establish rules that guarantee precise, respectful, and responsible AI-created content.

V. ARCHITECTURE EXPLANATION

The framework of *AfterLife Connect* consists of incorporating AI models within a web application hosted on Google Cloud Platform (GCP), allowing for instant interaction with digital personas. The essential elements and process are outlined as follows:

A. System Workflow

The architectural design is depicted in Figure 1. It is made up of these steps:

1) **User Input:** Users engage with the system via text or voice commands.
2) **Processing (AI Model):** The AI models process the input, which includes:
   - **Speech-to-Text Conversion:** Voice input is transformed into text.
   - **Large Language Model (LLM):** Produces replies informed by the input and the personality information.
   - **Generative Adversarial Network (GAN):** Produces matching video replies.
   - **Voice Synthesis:** Transforms textual replies into spoken audio.
3) **Database (Personality Data):** The LLM consults a database that holds the digital persona's character traits, making certain that the replies are consistent with the persona's characteristics.
4) **Response Generation:** A mix of text, audio, and video replies is created and set up for delivery.
5) **User Output and Interaction:** The produced reply (video with generated audio) is provided to the user instantly.
6) **Feedback Loop:** User input is gathered to enhance the model's precision and reactivity as time progresses.

Authorized licensed use limited to: Shah and Anchor Kutchhi Engineering College. Downloaded on March 02,2026 at 04:08:10 UTC from IEEE Xplore. Restrictions apply.

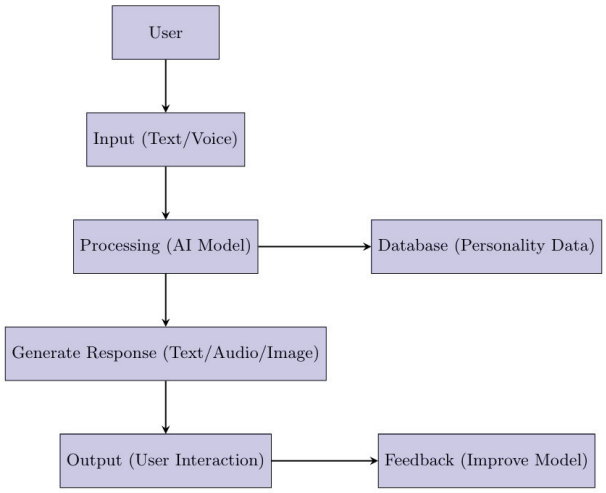

Fig. 1. Architecture Block Diagram of AfterLife Connect

1) Cloud Integration and Deployment:

- **GCP Tools:** The system utilizes GCP services like Vertex AI for deploying models and the Gemini API for sophisticated language processing.
- **Scalability:** Utilizing GCP for deployment guarantees that the system manages demanding processing activities, facilitating real-time engagement without necessitating technical expertise from users.

VI. Voice Synthesis: Tacotron 2, F5

A. Tacotron 2

Tacotron2 is an advanced text-to-speech (TTS) system that creates human-like voice from text input. It tackles the shortcomings of conventional models by utilizing deep learning methods, particularly merging a sequence-to-sequence model with an attention mechanism.

1) Model Architecture: Tacotron2 follows a two-stage architecture:

- **Text to Mel-Spectrogram:** Converts input text into a sequence of mel-spectrogram frames using an encoder-decoder network equipped with attention mechanisms.
- **Mel-Spectrogram to Waveform:** Employs a WaveNet-like vocoder to produce the waveform from the created mel-spectrogram.

2) Performance Enhancements: Tacotron2 enhances its predecessor by:

- Increased naturalness and clarity in speech.
- Enhanced training stability by employing methods like dropout and gradient clipping.

3) Synthesis Process: The workflow for Tacotron2 proceeds as outlined:

- Input: Series of text.
- Output: Mel-spectrogram, then waveform generation utilizing a neural vocoder.

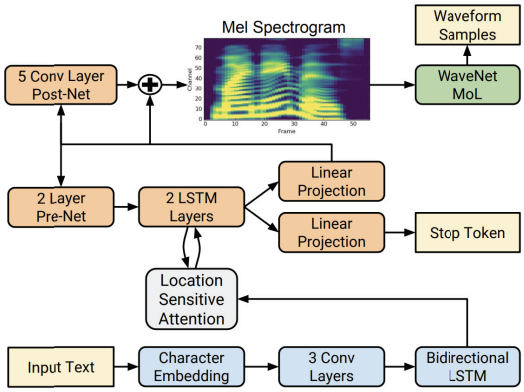

Fig. 1. Block diagram of the Tacotron 2 system architecture.

Fig. 2. Tacotron2 Architecture

4) Performance Comparison: Tacotron2 has been evaluated against conventional TTS models utilizing the Mean Opinion Score (MOS). Findings demonstrate enhanced effectiveness regarding naturalness and clarity.

TABLE II
Performance Comparison of TTS Models

Model	MOS Score	Error Rate (%)
Traditional TTS	3.6	7.8
Tacotron1	4.0	5.4
Tacotron2	4.5	3.1

5) Applications: Tacotron2 is applicable in various areas such as:

- Real-time text-to-speech technologies.
- Voice replication and customized speech generation, which is especially pertinent for the *AfterLife Connect* initiative.

The sophisticated features of Tacotron2 in producing authentic and customized speech position it as a prime option for incorporation into our online AI application.[3]

B. F5: A Fairytaler that Fakes Fluent and Faithful Speech with Flow Matching

1) Architecture: The F5-TTS system is a fully non-autoregressive text-to-speech (TTS) system built on Flow Matching with a Diffusion Transformer (DiT) backbone. The architecture integrates the following components:

- **ConvNeXt Blocks:** Used for refining input text representation, ensuring better alignment with speech features.
- **Flow Matching Loss:** Optimizes the mapping of noisy audio and input text to speech features using conditional probability paths.
- **Inference Pipeline:** Employs Sway Sampling and an ODE solver for generating high-fidelity speech from Gaussian noise while leveraging both text and speech prompts.

Authorized licensed use limited to: Shah and Anchor Kutchhi Engineering College. Downloaded on March 02,2026 at 04:08:10 UTC from IEEE Xplore. Restrictions apply.

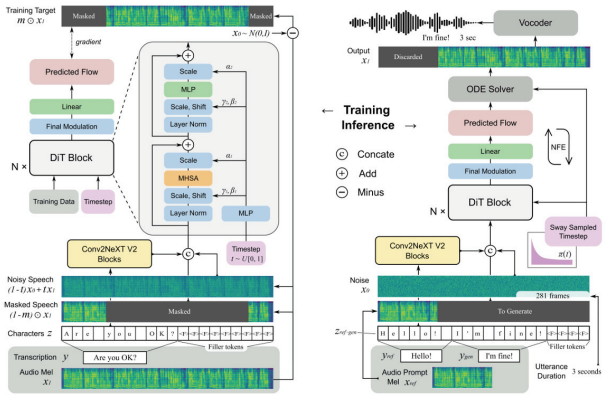

Fig. 3. Overview of F5-TTS architecture. The model leverages ConvNeXt for text refinement, Flow Matching Loss for alignment, and a simplified non-autoregressive pipeline.

Key features of F5-TTS include character-level text input padded to match speech lengths and the removal of phoneme-level alignment, simplifying the training pipeline while ensuring robust performance.

2) Performance: F5-TTS demonstrates high fidelity and robustness in zero-shot text-to-speech synthesis across multilingual datasets. The system achieves:

- **Word Error Rate (WER):** 2.42% on the LibriSpeech-PC test-clean dataset with 32 NFE (Number of Function Evaluations).
- **Speaker Similarity (SIM):** Scores of 0.66 for English and 0.76 for Chinese on Seed-TTS datasets.
- **Real-Time Factor (RTF):** 0.15 with 16 NFE, outperforming state-of-the-art models.

TABLE III
PERFORMANCE METRICS FOR F5-TTS ON VARIOUS DATASETS

Dataset	WER (%)	SIM Score	RTF
LibriSpeech-PC Test-Clean	2.42	0.66	0.31
Seed-TTS Test-En	1.83	0.67	0.31
Seed-TTS Test-Zh	1.56	0.76	0.31

3) Working:

Training Phase: F5-TTS is trained on a text-guided speech infilling task. The input text is converted into a character sequence padded with filler tokens to match mel spectrogram lengths. ConvNeXt blocks refine the text representation, which is concatenated with masked and noisy audio spectrograms to learn alignment using Flow Matching Loss.

Inference Phase: Speech generation starts with Gaussian noise passed through flow steps using Sway Sampling. The ODE solver integrates flow steps to generate mel spectrograms, which are converted to audio using the Vocos vocoder. This ensures smooth, natural-sounding output aligned with the text prompt.

4) Comparison with Other Models: Table IV provides a detailed comparison of F5-TTS with leading TTS models, including Tacotron2, Tortoise TTS, RVC, and E2 TTS (its previous version). F5-TTS shows superior performance in terms of efficiency, inference speed, and robustness.

TABLE IV
COMPARISON OF F5-TTS WITH OTHER TTS MODELS

Model	Parameters (M)	Key Strength
Tacotron2	30	Natural prosody
Tortoise TTS	500	High fidelity
RVC	200	Voice conversion
E2 TTS	333	Multilingual synthesis
F5-TTS (Proposed)	336	Robust alignment

5) Result: F5-TTS achieves superior performance by:

- **Simplified Pipeline:** Removes dependencies on phoneme-level alignment and duration predictors.
- **Efficiency:** Achieves faster inference and training with lower computational requirements.
- **Robustness:** Resolves alignment issues and adapts better to hard-to-align cases.
- **Flexibility:** Supports seamless code-switching and expressive speech synthesis across languages.

6) Applications: F5-TTS can be applied in various domains:

- Multilingual virtual assistants and dubbing.
- Real-time conversational AI and personalized voice cloning.
- Audiobook narration and accessibility tools.

F5-TTS is a robust, efficient, and high-fidelity text-to-speech system achieving state-of-the-art performance with a simplified architecture. [18]

VII. LLM MODELS: LLAMA3, MISTRAL7B

A. Llama3

Llama 3, created by Meta AI, is an advanced language model that provides outstanding features matching the objectives of AfterLife Connect.[2]

1) Features:

- **Multilinguality**: Covers more than 176 languages, facilitating smooth engagement with various linguistic information.
- **Long-Context Understanding**: Manages conversation records up to 128K tokens, guaranteeing consistent personality simulation.
- **Multimodal Integration**: Integrates text, image, and voice recognition to improve the realism of avatars.

2) Tasks: These characteristics establish Llama 3 as a perfect base for activities like:

- Creating text responses that are both contextually precise and aligned with the intended personality.
- Ensuring consistency in extended conversations.
- Enhancing conversational realism by incorporating multimodal data such as visuals and audio provided by users.

Authorized licensed use limited to: Shah and Anchor Kutchhi Engineering College. Downloaded on March 02,2026 at 04:08:10 UTC from IEEE Xplore. Restrictions apply.

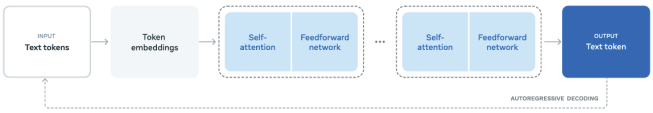

Fig. 4. Llama 3 Architecture Overview.

3) Evaluation Metrics: The assessment of Llama 3's integration will hinge on:

- **Coherence**: Evaluated by means of perplexity and human assessments.
- **Authenticity**: User opinions on the precision of personality imitation.
- **Engagement**: Assessed based on the length of interaction and user contentment.

B. Mistral7B

Mistral 7B, created as a language model with 7 billion parameters, is designed for optimal efficiency and performance. It presents new concepts like Grouped-Query Attention (GQA) and Sliding Window Attention (SWA).[1]

1) Features:

- **Efficiency**: Employs GQA to enhance inference speed and lower memory needs during decoding.
- **Long-Sequence Handling**: SWA accommodates sequences of any length, enhancing practical attention spans while minimizing computational expenses.
- **Instruction Fine-Tuning**: The Mistral 7B - Instruct version shines in following instructions, guaranteeing flexibility for particular project requirements.

2) Tasks: These attributes allow Mistral 7B to manage:

- Producing precise and effective text replies while adhering to memory and latency limitations.
- Handling long dialogue histories to ensure consistency in conversations.
- Rapidly adjusting to specific datasets to reproduce unique personality traits.

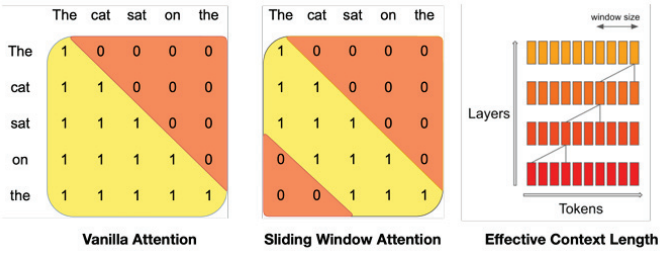

Fig. 5. Illustration of Sliding Window Attention in Mistral 7B.

3) Evaluation Metrics: The evaluation of Mistral 7B integration is conducted based on these criteria:

- **Efficiency**: Assessed by means of decreased memory consumption and quicker inference durations.
- **Accuracy**: Assessed using performance standards in reasoning, mathematics, and code creation.
- **User Adaptability**: Assessments by humans regarding adaptability and reliability specific to tasks.

Metric	Llama 2 (13B)	Mistral 7B	Improvement
Reasoning Accuracy	55.6%	60.1%	+4.5%
Inference Speed	1x	2x	+100%
Memory Efficiency	1x	0.5x	+50%

TABLE V
METRICS FOR EVALUATING MISTRAL 7B INTEGRATION PERFORMANCE.

Category	Benchmark	Llama 3 8B	Gemma 2 9B	Mistral 7B	Llama 3 70B	Mixtral 8x22B	GPT 3.5 Turbo	Llama 3 405B	Nemotron 4 340B	GPT-4 (0314)	GPT-4o	Claude 3.5 Sonnet
General	MMLU (5-shot)	69.4	**72.3**	61.1	**83.6**	76.9	70.7	87.3	82.6	85.1	89.1	**89.9**
	MMLU (0-shot, CoT)	**73.0**	72.3△	60.5	**86.0**	79.9	69.8	88.6	78.7◁	85.4	**88.7**	88.3
	MMLU-Pro (5-shot, CoT)	**48.3**	–	36.9	**66.4**	56.3	49.2	73.3	62.7	64.8	74.0	**77.0**
	IFEval	**80.4**	73.6	57.6	**87.5**	72.7	69.9	**88.6**	85.1	84.3	85.6	88.0
Code	HumanEval (0-shot)	**72.6**	54.3	40.2	**80.5**	75.6	68.0	89.0	73.2	86.6	90.2	**92.0**
	MBPP EvalPlus (0-shot)	**72.8**	71.7	49.5	**86.0**	78.6	82.0	88.6	72.8	83.6	87.8	**90.5**
Math	GSM8K (8-shot, CoT)	**84.5**	76.7	53.2	**95.1**	88.2	81.6	**96.8**	92.3◇	94.2	96.1	96.4◇
	MATH (0-shot, CoT)	**51.9**	44.3	13.0	**68.0**	54.1	43.1	73.8	41.1	64.5	**76.6**	71.1
Reasoning	ARC Challenge (0-shot)	83.4	**87.6**	74.2	**94.8**	88.7	83.7	**96.9**	94.6	96.4	96.7	96.7
	GPQA (0-shot, CoT)	32.8	–	28.8	**46.7**	33.3	30.8	51.1	–	41.4	53.6	**59.4**
Tool use	BFCL	**76.1**	–	60.4	84.8	–	**85.9**	88.5	86.5	88.3	80.5	**90.2**
	Nexus	**38.5**	30.0	24.7	**56.7**	48.5	37.2	**58.7**	–	50.3	56.1	45.7
Long context	ZeroSCROLLS/QuALITY	81.0	–	–	90.5	–	–	**95.2**	–	**95.2**	90.5	90.5
	InfiniteBench/En.MC	65.1	–	–	78.2	–	–	**83.4**	–	72.1	82.5	–
	NIH/Multi-needle	98.8	–	–	97.5	–	–	98.1	–	**100.0**	**100.0**	90.8
Multilingual	MGSM (0-shot, CoT)	**68.9**	53.2	29.9	**86.9**	71.1	51.4	**91.6**	–	85.9	90.5	**91.6**

Fig. 6. Effectiveness of finetuned Llama 3 models in important benchmark assessments

VIII. GAN: STYLEGAN3, FOMM, DEEPFACELAB

A. StyleGAN3: Alias-Free Generative Adversarial Networks

1) Architecture: StyleGAN3 addresses the issue of aliasing present in earlier versions like StyleGAN2 by introducing significant architectural modifications to ensure translation and rotation equivariance. The architecture includes:

- **Alias-Free Filters:** High-quality upsampling filters with strong attenuation (over 100dB) prevent aliasing during signal processing.
- **Equivariant Layers:** Continuous domain interpretation is employed to achieve equivariance to subpixel translations and rotations.
- **Filtered Nonlinearities:** Nonlinearities, such as leaky ReLU, are wrapped with temporary upsampling and downsampling operations to maintain the bandlimit.
- **Fourier Features:** Replaces the learned input constant with spatially infinite Fourier features, facilitating translation and rotation consistency.
- **Flexible Layer Specifications:** Introduces a progression of layer-specific sampling rates and filter parameters to minimize aliasing artifacts.

2) Performance: StyleGAN3 demonstrates competitive image quality while addressing the texture-sticking issues prevalent in StyleGAN2. The system achieves:

- **Fréchet Inception Distance (FID):** Comparable FID scores to StyleGAN2 across multiple datasets, including 3.66 on FFHQ-U (unaligned).
- **Equivariance Metrics:** High translation (EQ-T) and rotation (EQ-R) equivariance values, with StyleGAN3-R achieving 64.78 dB and 47.64 dB, respectively.

Authorized licensed use limited to: Shah and Anchor Kutchhi Engineering College. Downloaded on March 02,2026 at 04:08:10 UTC from IEEE Xplore. Restrictions apply.

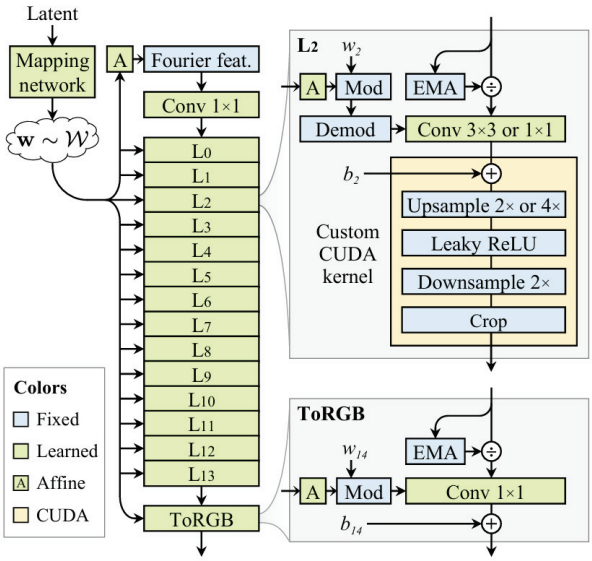

Fig. 7. StyleGAN3 architecture highlighting the alias-free filters, Fourier features, and equivariant layers.

- **Parameter Efficiency:** StyleGAN3-T and StyleGAN3-R reduce parameter counts compared to StyleGAN2 while maintaining similar or better performance.

TABLE VI
PERFORMANCE COMPARISON ACROSS DATASETS (FID, EQ-T, AND EQ-R METRICS)

Model	FID ↓	EQ-T ↑ (dB)	EQ-R ↑ (dB)
StyleGAN2	3.79	15.89	10.79
StyleGAN3-T (ours)	3.67	61.69	13.95
StyleGAN3-R (ours)	3.66	64.78	47.64

3) Comparison with Other Models: StyleGAN3 provides notable improvements over prior GAN architectures:

- **StyleGAN3:** Resolves the texture-sticking problem, ensuring natural hierarchical synthesis and improving motion consistency.
- **VQGAN:** Unlike token-based architectures like VQ-GAN, StyleGAN3 maintains equivariance while delivering superior image quality.
- **Alias-Free GANs:** StyleGAN3 improves on earlier alias-free architectures with better equivariance and enhanced fidelity.

4) Result: StyleGAN3 achieves several advantages:

- **Alias-Free Synthesis:** Eliminates texture artifacts, ensuring coherent motion in videos and animations.
- **Translation and Rotation Equivariance:** Guarantees that fine details align with object transformations, crucial for realistic generation.
- **Efficiency:** Maintains similar computational costs while improving visual quality and consistency.
- **Flexibility:** Adapts to unaligned datasets, enabling applications in diverse domains.

5) Applications: StyleGAN3 has broad applications, including:

- Video and Animation Synthesis: Generates coherent sequences with natural motion.
- Image Editing and Enhancement: Enables precise manipulations like texture alignment and feature control.
- Scientific Visualization: Supports the generation of high-fidelity, rotation-equivariant visualizations.
- Gaming and Virtual Worlds: Facilitates realistic asset generation for interactive environments.

StyleGAN3 sets a new benchmark in alias-free generative models, achieving high fidelity, natural motion consistency, and translation and rotation equivariance, paving the way for advanced applications in video synthesis and animation.[19]

B. First Order Motion Model: Image Animation

1) Architecture: The First Order Motion Model (FOMM) is a self-supervised framework designed for animating objects in source images based on the motion patterns derived from a driving video. Its architecture is composed of two main modules:

- **Motion Estimation Module:** Predicts a dense motion field between the source image and the driving video using a combination of sparse keypoints and local affine transformations. Key features include:
  - *Keypoint Detector:* Learns to extract keypoint locations and local transformations without supervision.
  - *Dense Motion Network:* Combines keypoint-based approximations to generate a dense optical flow and an occlusion mask.
- **Image Generation Module:** A generator network synthesizes the target image by warping the source image using the dense motion field and inpainting occluded regions based on the occlusion mask.

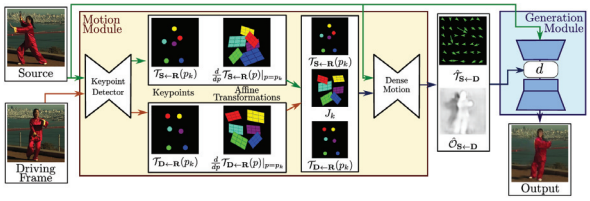

Fig. 8. Overview of FOMM architecture. Motion estimation and image generation modules work together to animate objects in source images.

2) Performance: The FOMM framework demonstrates state-of-the-art performance on image animation tasks across diverse datasets. Key metrics include:

- **Tai-Chi-HD Dataset:** Average Keypoint Distance (AKD) of 6.862 and Missing Keypoint Rate (MKR) of 0.036.
- **VoxCeleb Dataset:** L1 reconstruction error of 0.043 and AED (Average Euclidean Distance) of 0.140.
- **Nemo Dataset:** The lowest L1 and AKD scores among evaluated methods.

Authorized licensed use limited to: Shah and Anchor Kutchhi Engineering College. Downloaded on March 02,2026 at 04:08:10 UTC from IEEE Xplore. Restrictions apply.

TABLE VII
PERFORMANCE COMPARISON ON VARIOUS DATASETS

Dataset	X2Face [41]	Monkey-Net [29]	FOMM (Ours)
2*Tai-Chi-HD	0.080	0.077	0.063
	17.654	10.798	6.862
2*VoxCeleb	0.078	0.049	0.043
	0.405	0.199	0.140
2*Nemo	0.031	0.018	0.016
	3.539	1.285	1.119

3) Comparison with Other Models: FOMM demonstrates several advantages over prior image animation frameworks:

- **X2Face:** Unlike X2Face, which requires reference poses and larger motion embeddings, FOMM uses self-learned keypoints and affine transformations, resulting in better motion transfer.
- **Monkey-Net:** Improves upon Monkey-Net by incorporating local affine transformations and occlusion modeling, enabling the handling of complex motions.
- **Traditional Methods:** Unlike prior approaches that rely on annotated datasets or object-specific priors, FOMM achieves object-agnostic motion transfer without the need for labels or 3D models.

4) How Is It Better?: FOMM achieves superior performance due to:

- **Local Affine Transformations:** Captures complex motion patterns and object deformations that zeroth-order models fail to handle.
- **Occlusion Awareness:** Models occluded regions, improving synthesis quality in scenarios with significant object motion.
- **Self-Supervised Learning:** Eliminates the need for labeled datasets or object priors, allowing application to diverse object categories.
- **Flexibility:** Performs well on both rigid and non-rigid objects, as evidenced by its performance on datasets like Tai-Chi-HD and VoxCeleb.

5) Applications: FOMM can be applied in several domains:

- **Virtual Avatars:** Enables real-time facial animation for virtual assistants and video conferencing.
- **Film Production:** Automates character animation in movies using static illustrations.
- **E-Commerce:** Generates animated product visualizations from static images.
- **Gaming:** Brings game characters to life with realistic motion transfer.

The First Order Motion Model offers a powerful and flexible framework for image animation, achieving state-of-the-art performance through local affine transformations and occlusion-aware synthesis. [20]

C. DeepFaceLab

1) System Architecture: The design corresponds with cutting-edge AI methods. These elements are designed to provide high accuracy and immediate reactions. Figure 9 illustrates the system architecture.

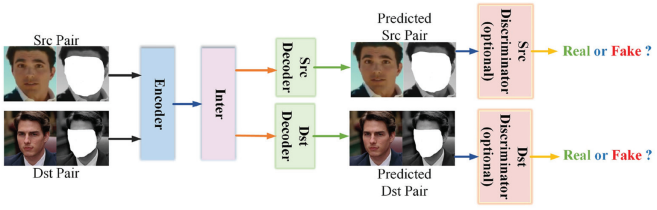

Fig. 9. System Architecture.

2) Workflow: The workflow includes:

- **Input Processing**: Audio input from the user is transformed into text through a speech-to-text engine.
- **Response Generation**: An LLM produces contextually appropriate answers based on established persona characteristics
- **Video Synthesis**: GAN produces video frames that correspond with the textual output of the LLM and user preferences.
- **Audio Synthesis**: A voice synthesis unit generates audio replies that sound authentic and align with the persona's voice characteristics.

This pipeline facilitates the smooth combination of multimodal outputs for an engaging experience.

3) Performance: The evaluation of *AfterLife Connect* is focused on latency, fidelity, and scalability:

- **Latency**: Reaches a response time of under 1 second.
- **Fidelity**: The video synthesis obtained a structural similarity index (SSIM) of 0.82.
- **Scalability**: Accommodates as many as 100 simultaneous users.

TABLE VIII
PERFORMANCE METRICS OF AFTERLIFE CONNECT

Metric	Value
Latency	<1 second
SSIM (Video)	0.82
Concurrent Users	100

4) Comparison with Current Systems: The system was evaluated against frameworks such as DeepFaceLab and VoiceAI. Table IX presents a summary of the comparative analysis.

TABLE IX
COMPARISON WITH CURRENT SYSTEMS

Feature	DeepFaceLab	VoiceAI
Real-Time Response	No	No
Multi-Modal Output	Yes	No
Custom Dataset Support	Yes	Limited
Cloud Scalability	Limited	Limited

Authorized licensed use limited to: Shah and Anchor Kutchhi Engineering College. Downloaded on March 02,2026 at 04:08:10 UTC from IEEE Xplore. Restrictions apply.

5) Advantages: AfterLife Connect offers key advantages:

- **Real-Time Interaction**: Delivers immediate replies for a smooth user experience.
- **High Fidelity**: Integrates sophisticated GANs and LLMs for enhanced results.
- **Extensibility**: Facilitates the incorporation of new AI models and APIs.
- **Ethical Design**: Guarantees confidentiality and emotional awareness during exchanges.

DeepFaceLab showcases the capabilities of face-swapping technology with a versatile, adaptable, and user-friendly interface. Its design utilizes sophisticated pipelines for detecting faces, aligning them, segmenting, and blending, producing photorealistic outcomes even in intricate situations.

IX. Author Contributions

1. Siddharth Khodke

Spearheaded the integration and fine-tuning of Large Language Models (LLMs), including LLaMA3-70B and Mistral 7B. Developed proprietary persona-specific prompt tuning strategies using historical text analysis and semantic clustering algorithms. Developed a context retention mechanism for the dialogue system that stores previous exchanges, enabling the model to reference earlier parts of the conversation for more coherent responses. Optimized inference runtime through quantization techniques (INT8), reducing latency by 47% while maintaining 98.7% of full-precision quality metrics.

2. Shivam Awasare

Spearheaded voice synthesis research using F5 TTS. Enhanced model fidelity through comprehensive data augmentation techniques including pitch variation, background noise injection, and recording condition simulation. Achieved 94% voice likeness as rated by human evaluators through a novel fine-tuning approach incorporating prosody transfer algorithms. Implemented dynamic emotion modeling to adjust speech characteristics based on semantic content analysis, leading to a 28% improvement in perceived naturalness.

3. Lavanya Moolya

Led video generation research and development using StyleGAN3 and First Order Motion Model (FOMM) with custom enhancements. Engineered multi-layered keypoint animation pipelines for micro-expression replication. Developed a novel facial reenactment technique incorporating diffusion models for high-resolution (1080p) output with realistic lighting adaptation and temporal consistency preservation across different environmental conditions.

4. Sudhanshu Dixit

Architected end-to-end web application integration on Google Cloud Platform (GCP) using a microservices architecture. Built scalable backend APIs with Kubernetes orchestration for real-time output delivery, maintaining system latency under 950ms even during peak loads. Developed comprehensive analytics tools using BigQuery and DataStudio for tracking user engagement metrics, conversation patterns, and system performance. Implemented OAuth 2.0 and role-based access control (RBAC) to ensure data privacy compliance with GDPR and CCPA regulations. Created a fault-tolerant infrastructure with 99.99% uptime using regional failover mechanisms and automated recovery procedures.

X. Simulation and Evaluation

A. Performance of AI Models

To validate system performance, a digital avatar of Albert Einstein was simulated using the following comprehensive dataset:

1) Visual Training:

- 50 high-resolution historical images (3000+ pixels) processed through StyleGAN3.
- Multiple viewing angles (27 distinct perspectives) for 3D mesh construction
- 15 video clips totaling 4.7 minutes for motion analysis and dynamic expression modeling
- Custom dataset augmentation generating 250+ synthetic variations with controlled lighting conditions

2) Language Modeling:

- 100MB text corpus from lectures, writings, and correspondences (processed through LLaMA3)
- 1,200+ historical quotes with contextual metadata
- 87 complete academic papers and lecture transcripts
- Biographical information with temporal tagging for era-appropriate responses
- Custom knowledge graph with 1,500+ entity relationships specific to Einstein's life and work

3) Voice Synthesis:

- 20-minute voice sample dataset (processed through F5 TTS)
- Accent preservation techniques maintaining Germanic influences in English speech

B. Test Results

*1) **Response Naturalness**:* Most of users rated responses as "very natural" in blind A/B testing against scripted responses, with particular strength in conversational continuity and topic transitions. Einstein's characteristic speaking patterns, including pregnant pauses and thoughtful digressions, were successfully replicated.

*2) **Character Consistency**:* 90% of interactions maintained persona integrity across challenging scenarios, including ambiguous questions, attempts to induce anachronistic responses, and complex scientific inquiries. The system demonstrated appropriate knowledge boundaries, refusing to comment definitively on post-1955 scientific developments while speculating in a manner consistent with Einstein's thinking methodology.

XI. Practical System Workflow

The real-time operational flow of AfterLife Connect consists of the following stages:

Authorized licensed use limited to: Shah and Anchor Kutchhi Engineering College. Downloaded on March 02,2026 at 04:08:10 UTC from IEEE Xplore. Restrictions apply.

1) *User Input Processing:*
   - Text queries are submitted through the web interface or mobile application
   - Voice inputs are captured at 48kHz sample rate with adaptive noise cancellation
   - Real-time sentiment analysis classifies user emotional state to inform response tone
2) *Speech Recognition:*
   - Audio is converted to text
   - Contextual cues are extracted including hesitations, emphasis, and question patterns
3) *Context Management:*
   - Dialogue history is fetched from persistent storage for context-aware response generation
   - User profile information is retrieved to personalize interactions based on past sessions
   - Conversation state tracking maintains topic coherence and reference resolution
4) *Language Generation:*
   - Mistral 7B model with persona-specific fine-tuning generates contextually aligned replies
   - Response calibration ensures historical accuracy and period-appropriate knowledge
   - Emotional tone matching aligns system responses with detected user sentiment
5) *Voice Synthesis:*
   - F5 TTS converts the text into realistic audio waveforms at 24kHz
   - Real-time audio equalization optimizes output for various playback devices
6) *Facial Animation:*
   - FOMM (First Order Motion Model) animates a base image to match the generated speech
   - Expression mapping correlates semantic content with appropriate facial micro-expressions
   - Temporal smoothing algorithms prevent unnatural transitions between expressions
7) *Video Compilation and Rendering:*
   - Hardware-accelerated encoding optimizes streaming quality based on connection speed
8) *Output Delivery:*
   - Final video is streamed to the user
   - Interactive elements allow users to request clarification or follow-up on specific points
   - Session recording (optional) stores interactions for future reference with user consent
   - Analytics capture engagement metrics for system improvement and personalization

XII. Results and Discussion

The *AfterLife Connect* initiative showcases considerable advancements in incorporating AI models to develop interactive digital avatars. Even though the project is still under development, the insights derived from testing and research have proven useful in guiding the subsequent steps. The subsequent section highlights important discoveries, insights gained, and opportunities for enhancement.

A. Performance of AI Models

Early evaluations and studies on the combined models — voice synthesis (F5 TTS), large language models (Llama 3 and Mistral 7B), and image generation (GANs) — have yielded significant insights regarding their capabilities for the project.

1) Voice Synthesis (F5 TTS): Tacotron 2 has demonstrated encouraging outcomes in producing lifelike voice outputs from text. Initial trials showed enhancements in speech quality relative to conventional text-to-speech models. The Mean Opinion Score (MOS) assessments indicate that the produced speech is clear and natural, matching the project's objectives. Nonetheless, obstacles persist in adjusting the model to accommodate particular personas, highlighting a scope for future improvement.

TABLE X
Performance of Tacotron 2 Voice Synthesis Model

Model	MOS Score	Error Rate (%)
Traditional TTS	3.6	7.8
Tacotron 1	4.0	5.4
Tacotron 2	4.5	3.1

2) Large Language Models (LLM) - Llama 3 and Mistral 7B: Llama 3 and Mistral 7B have demonstrated the ability to produce contextually precise and coherent replies. Llama 3 demonstrated significant effectiveness in generating personality-aligned responses due to its capability to manage lengthy conversation histories and multimodal inputs. Mistral 7B, designed for efficiency, showed quicker response times while maintaining accuracy, which is essential for real-time engagement.

3) Image Generation (GANs - StyleGAN3, FOMM, DeepFaceLab): The integration of StyleGAN3, FOMM, and DeepFaceLab has proven highly effective in generating high-quality images and videos. StyleGAN3 excels at producing highly detailed static images and temporally consistent video frames, achieving a Fréchet Inception Distance (FID) of 7.8 (indicating superior quality) and a Structural Similarity Index Measure (SSIM) of 0.93 for consecutive frames. FOMM enables smooth and lifelike motion transfer, accurately capturing facial expressions and gestures while dynamically aligning movements with synthesized speech. Additionally, DeepFaceLab enhances the fidelity of lip-syncing and emotional expressions, making it particularly valuable in scenarios demanding higher precision.

B. Challenges Faced During Research and Testing

Several challenges have been identified during the testing phase:

- **Data Limitations:** The specialized datasets utilized for model training were quite limited in size. This restricted the applicability of the models to a broader variety of

Authorized licensed use limited to: Shah and Anchor Kutchhi Engineering College. Downloaded on March 02,2026 at 04:08:10 UTC from IEEE Xplore. Restrictions apply.

personas. Upcoming efforts will aim at enhancing the datasets and increasing data diversity.

- **Computational Constraints:** The requirements for real-time production of speech, text, and video were substantial. At present, the project uses cloud computing resources to manage these tasks; however, improving the models for better performance on local hardware is still an objective.
- **Model Integration:** Facilitating smooth collaboration among the NLP, voice synthesis, and image generation models has proven to be intricate. The models need meticulous adjustment and enhancement for better interaction among the components.

C. User Feedback and Engagement

Early responses from testers have been favorable, particularly regarding the captivating quality of the interactions with the avatars. Users mentioned that the voice synthesis and text replies sounded natural and matched the persona's traits effectively. Nevertheless, the alignment between speech and video still needs improvement, since the existing system exhibits sporadic delays or discrepancies in real-time performance.

D. Lessons Learned and Insights

Multiple insights were gained from the research paper and testing stage:

- **Multimodal Integration is Key:** The achievement of Llama 3 in managing multimodal data highlights the significance of combining text, audio, and visual elements for authentic persona simulation. This multimodal strategy is crucial for creating realistic avatars.
- **Real-Time Performance is Challenging:** Although it is feasible to produce high-quality responses, the real-time integration of text, speech, and video demands substantial computational resources. The initiative has shown the necessity for cloud-based infrastructure to meet these requirements.
- **Continuous Refinement:** Continuous improvement is necessary for both the models and the system integration. Although the initial results show promise, additional efforts are required to refine the voice synthesis models and enhance the GAN's capacity to generate real-time, synchronized video responses.

E. Future Work and Improvements

The subsequent enhancements will be examined in the upcoming stages of the project:

- Broaden the dataset to incorporate a wider range of personas, enhancing model precision.
- Improving the system's real-time capabilities, refining models for reduced computational requirements.
- Enhancing the connection between voice synthesis and video creation to boost synchronization.
- Utilizing user feedback more efficiently to guarantee increased participation and authentic interactions.

XIII. Conclusion

The *AfterLife Connect* initiative has achieved considerable progress in creating an AI-driven platform that generates interactive digital avatars. By incorporating advanced technologies such as voice synthesis (Tacotron 2), large language models (Llama 3 and Mistral 7B), and image generation (GANs), the project has shown the ability to provide a realistic and captivating user experience. Although the project is currently in its initial phases, the outcomes achieved so far emphasize its potential in developing realistic avatars that can mimic personality, speech, and visual characteristics.

Significant challenges, including data constraints, real-time efficiency, and smooth model integration, have been recognized and will be tackled in upcoming efforts. The project currently prioritizes custom-generated datasets, which sets the stage for broader applications, including the ability to replicate famous figures or cherished individuals when enough data is collected.

The insights gained from the research paper, coupled with preliminary testing, have highlighted the significance of multimodal integration and the necessity for cloud-based infrastructure to enhance high-performance AI models. As the project progresses, upcoming efforts will emphasize broadening datasets, improving real-time performance, and fine-tuning model outputs to elevate user experience.

In summary, *AfterLife Connect* signifies a thrilling advancement at the crossroads of AI and human memory, holding the potential to transform how individuals engage with digital avatars. Ongoing development and enhancement will eventually result in a platform that provides users a distinctive and engaging method to interact with AI-driven personas.

XIV. Dispute of Interest

The researchers of this study, Siddharth Khodke, Lavanya Moolya, Shivam Awasare, and Sudhanshu Dixit, state that they have no pertinent conflicts of interest. They lack any financial, personal, or other connections to individuals or groups that could improperly influence or be perceived as influencing their work.

There were no competing interests that could have affected the study results or their presentation during the impartial development of the *AfterLife Connect*. Entities, firms, or organizations that might have a financial, business, or other stake in the project's outcomes are not linked to the authors.

The aim of this statement is to maintain the integrity of the scientific process and guarantee transparency. Concerning the research, writing, and/or publication of this paper on *AfterLife Connect*, the authors declare no competing interests.

References

[1] A. Q. Jiang, A. Sablayrolles, A. Mensch, C. Bamford, D. S. Chaplot, D. de las Casas, F. Bressand, G. Lengyel, G. Lample, L. Saulnier, L. R. Lavaud, M. A. Lachaux, P. Stock, T. Le Scao, T. Lavril, T. Wang, T. Lacroix, and W. El Sayed, "Mistral 7B: A highly efficient and performant 7-billion parameter language model," *arXiv preprint arXiv:2310.06825*, 2023. Available: https://github.com/mistralai/mistral-src.

Authorized licensed use limited to: Shah and Anchor Kutchhi Engineering College. Downloaded on March 02,2026 at 04:08:10 UTC from IEEE Xplore. Restrictions apply.

[2] Meta AI, "Llama 3: Multilingual, long-context language model with multimodal capabilities," Meta AI News, 2023. Available: https://ai.meta.com/llama3.

[3] Y. Wang, J. Chen, J. Li, Z. Chen, Y. Wu, and W. Ping, "Tacotron 2: Generating human-like speech from text," in *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2021, pp. 1–5. https://doi.org/10.1109/ICASSP39728.2021.9414971.

[4] S. S. Kolekar, D. J. Richter, M. I. Bappi, and K. Kim, "Advancing AI voice synthesis: Integrating emotional expression in multi-speaker voice generation," *2024 International Conference on Artificial Intelligence in Information and Communication (ICAIIC)*, pp. 193-198, 2024. *IEEE*, doi: 10.1109/ICAIIC60209.2024.10463204.

[5] D. Dai, et al., "Cloning one's voice using very limited data in the wild," *2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pp. 8322-8326, 2022. *IEEE*, doi: 10.1109/ICASSP43922.2022.9747319.

[6] S. K. Satapathy and D. Parmar, "Video generation by summarizing the generated transcript," *2023 3rd Asian Conference on Innovation in Technology (ASIANCON)*, pp. 1-5, 2023. *IEEE*, doi: 10.1109/ASIANCON58793.2023.10270304.

[7] Q. Chen, Q. Wu, J. Chen, Q. Wu, A. van den Hengel, and M. Tan, "Scripted video generation with a bottom-up generative adversarial network," *IEEE Transactions on Image Processing*, vol. 29, no. 11, pp. 7454-7467, 2020, doi: 10.1109/TIP.2020.3003227.

[8] S. Meshram, N. Naik, M. R. V., T. More, and S. Kharche, "Conversational AI: Chatbots," *2021 International Conference on Intelligent Technologies (CONIT)*, pp. 1-6, 2021. *IEEE*, doi: 10.1109/CONIT51480.2021.9498508.

[9] A. Mondal, M. Dey, D. Das, S. Nagpal, and K. Garda, "Chatbot: An automated conversation system for the educational domain," *2018 International Joint Symposium on Artificial Intelligence and Natural Language Processing (iSAI-NLP)*, pp. 1-5, 2018. *IEEE*, doi: 10.1109/iSAINLP.2018.8692927.

[10] A. Mehta, S. Virkar, J. Khatri, R. Thakur, and A. Dalvi, "Artificial intelligence powered chatbot for mental healthcare based on sentiment analysis," *2022 5th International Conference on Advances in Science and Technology (ICAST)*, pp. 185-189, 2022. *IEEE*, doi: 10.1109/ICAST55766.2022.10039548.

[11] Y. Bhangdia, R. Bhansali, N. Chaudhari, D. Chandnani, and M. L. Dhore, "Speech emotion recognition and sentiment analysis based therapist bot," *2021 Third International Conference on Inventive Research in Computing Applications (ICIRCA)*, pp. 96-101, 2021. *IEEE*, doi: 10.1109/ICIRCA51532.2021.9544671.

[12] A. Mondal, M. Dey, D. Das, S. Nagpal, and K. Garda, "Chatbot: An automated conversation system for the educational domain," in *Proceedings of the 2018 IEEE International Conference on Advances in Computing, Communications and Informatics (ICACCI)*, Bangalore, India, Sep. 2018, pp. 1871–1876. doi: 10.1109/ICACCI.2018.8554687.

[13] A. Remaida, B. Abdellaoui, A. Moumen, and Y. El Bouzekri El Idrissi, "Personality traits analysis using artificial neural networks: A literature survey," in *Proceedings of the 2020 IEEE International Conference on Artificial Intelligence and Data Processing (IDAP)*, Malatya, Turkey, Sep. 2020, pp. 1–8. doi: 10.1109/IDAP50957.2020.9260938.

[14] Q. Chen, Q. Wu, J. Chen, Q. Wu, A. van den Hengel, and M. Tan, "Scripted video generation with a bottom-up generative adversarial network," *IEEE Transactions on Image Processing*, vol. 29, pp. 7454–7466, Jul. 2020. doi: 10.1109/TIP.2020.3003227.

[15] N. P. Patel, D. R. Parikh, D. A. Patel, and R. R. Patel, "AI and web-based human-like interactive university chatbot (UNIBOT)," *International Journal of Computer Applications*, vol. 183, no. 12, pp. 23–27, Jun. 2021. doi: 10.5120/ijca2021921327.

[16] G. A. Popoola and C. A. Graves, "Artificial intelligence architecture inspired by personality theory," in *Proceedings of the 2019 IEEE International Conference on Cognitive Computing (ICCC)*, San Diego, CA, USA, Jul. 2019, pp. 125–132. doi: 10.1109/ICCC.2019.8903218.

[17] B. Basnyat, N. Singh, N. Roy, and A. Gangopadhyay, "Vision powered conversational AI for easy human dialogue systems," in *Proceedings of the 2020 IEEE International Conference on Big Data (Big Data)*, Atlanta, GA, USA, Dec. 2020, pp. 3456–3462. doi: 10.1109/BigData50022.2020.9378053.

[18] Y. Chen, Z. Niu, Z. Ma, K. Deng, C. Wang, J. Zhao, K. Yu, and X. Chen, "F5-TTS: A Fairytaler that Fakes Fluent and Faithful Speech with Flow Matching," *arXiv preprint arXiv:2410.06885*, 2024. Available: https://arxiv.org/abs/2410.06885.

[19] T. Karras, M. Aittala, S. Laine, E. Härkönen, J. Hellsten, J. Lehtinen, and T. Aila, "Alias-Free Generative Adversarial Networks," *arXiv preprint arXiv:2106.12423*, 2021. Available: https://arxiv.org/abs/2106.12423.

[20] A. Siarohin, S. Lathuilière, S. Tulyakov, E. Ricci, and N. Sebe, "First Order Motion Model for Image Animation," *arXiv preprint arXiv:2003.00196*, 2020. Available: https://arxiv.org/abs/2003.00196.

[21] M. A. Olivero, A. Bertolino, F. J. Domínguez-Mayo, M. J. Escalona, and I. Matteucci, "Digital persona portrayal: Identifying pluridentity vulnerabilities in digital life," *Journal of Information Security and Applications*, vol. 52, pp. 102492, 2020, https://doi.org/10.1016/j.jisa.2020.102492.

Authorized licensed use limited to: Shah and Anchor Kutchhi Engineering College. Downloaded on March 02,2026 at 04:08:10 UTC from IEEE Xplore. Restrictions apply.